

Integers And Rational Numbers

mathNexa

Name:

Date:

Success Criteria:

- Identify and plot integers on real-world scales
- Find absolute values and opposites of numbers
- Compare and order positive and negative rational numbers
- Solve real-world integer and rational number problems

1. An outdoor thermometer reads 4°C below zero on a winter morning. Which integer correctly represents this temperature in degrees Celsius?

- ☐ A 4 ☐ B 0 ☐ C -4 ☐ D -14

2. A research submarine is positioned 175 meters below sea level. What integer represents this depth relative to sea level?

Answer: _____

3. A transaction fee of $-\$4.75$ is charged to a bank account. What is the absolute value of $-\$4.75$, and what is its opposite value?

- ☐ A Absolute value: -4.75 , Opposite: -4.75 ☐ B Absolute value: -4.75 , Opposite: 4.75
- ☐ C Absolute value: 4.75 , Opposite: 4.75 ☐ D Absolute value: 4.75 , Opposite: -4.75

4. At midnight, the temperature was -6°C . By noon, the temperature rose to 5°C . What is the total change in temperature in degrees Celsius?

Answer: _____ $^{\circ}\text{C}$

5. Which inequality correctly compares the two rational numbers $-\frac{3}{4}$ and -0.8 ?

- ☐ A $-\frac{3}{4} > -0.8$ ☐ B $-0.8 > -\frac{3}{4}$
- ☐ C $-\frac{3}{4} = -0.8$ ☐ D $-\frac{3}{4} < -0.8$

6. What is the mixed number $-2\frac{1}{4}$ expressed as an equivalent decimal?

Answer: _____

7. Four laboratory freezers are set to -12.5°C , -8.2°C , -15.0°C , and -9.8°C . Which freezer has the coldest (lowest) temperature setting?

☐ A -15.0°C

☐ B -8.2°C

☐ C -9.8°C

☐ D -12.5°C

8. A student account started with a balance of \$15.00. During the week, there was a debit of $-\$22.50$ and a deposit of $+\$30.00$. What is the final account balance?

Answer: _____

9. Which list shows the rational numbers ordered correctly from least to greatest: $-1.5, -\frac{2}{3}, 0.4, -2.1, 1\frac{1}{2}$?

☐ A $-\frac{2}{3}, -1.5, -2.1, 0.4, 1\frac{1}{2}$

☐ B $-2.1, -1.5, -\frac{2}{3}, 0.4, 1\frac{1}{2}$

☐ C $-1.5, -2.1, -\frac{2}{3}, 0.4, 1\frac{1}{2}$

☐ D $-2.1, -1.5, 0.4, -\frac{2}{3}, 1\frac{1}{2}$

10. A drone flies at an elevation of $+142.5$ meters above sea level directly over a marine sensor anchored at -58.5 meters below sea level. What is the total vertical distance between the drone and the sensor?

Answer: _____

Answers

1. C

2. -175

3. C

4. 11

5. A

6. -2.25

7. A

8. \$22.50

9. B

10. 201 m